# Context-Aware Speed and Separation Monitoring for Human–Robot Ceiling-Panel Installation: An Exploratory Evaluation

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***Abstract*—Ceiling-panel installation is demanding overhead work that a collaborative robot could share by holding the panel while a worker fixes it. Today's safeguard, fixed distance zones, cannot tell whether the worker is approaching, working or moving away, so it either stops the robot more than needed or must be set very conservatively. This study asks whether a robot that recognises the task phase can keep the worker as safe as fixed-zone control while keeping the work moving. We built a physical prototype in which a hidden Markov model recognises the task phase from motion capture and a separate predictor flags fast movement toward the robot, with a fixed rule that neither learned part can ever make the robot faster than the distance-based limit allows. Participants completed the task under fixed, reactive and predictive control. The phase model recognised what unseen operators were doing well enough to run in real time, showing that task context can be read reliably on a construction-style task. On safety, the answer to the research question is encouraging: every controller stopped the robot whenever a person entered the inner boundary, close approaches were alike, and participants trusted the robot to stop and felt safer after working with it than they expected. On efficiency the answer is mixed: tasks took no longer, but the predictive robot paused more often, mainly because its fast-approach predictor switched on and off on noisy readings. The hypothesis is therefore partly supported. For industry, the result means task awareness can be added on top of an unchanged safety limit without weakening it, but it does not yet deliver fewer interruptions; that benefit depends on smoothing the predictor and validating the complete sensing and stopping chain, which this study defines.**

***Index Terms*—Human–robot collaboration, speed and separation monitoring, hidden Markov model, construction robotics, motion capture, human factors.**

## I. Introduction and Background

INSTALLING ceiling panels involves lifting, aligning and fixing panels overhead, work that is physically demanding and usually done by people alone or with lifting aids. Collaborative robots could take over the lifting and holding [1]; the arrangement adopted in this study is for a robot to hold the panel while the worker provides judgement and fine hand skills (Fig. 1a). The worker then moves around a robot that is carrying a panel, the panel enlarges the space the robot occupies, and unnecessary robot stops can break the flow of work.

Speed and separation monitoring (SSM) keeps enough distance between a person and a robot, and slows or stops the robot in time [2]–[4]. The simplest form uses fixed distance thresholds. These are easy to understand, but they treat a person walking toward the robot, standing still and walking away in the same way. Knowing the task phase could help the robot avoid unnecessary stops, as long as a wrongly recognised phase can never loosen the safety limit. Our controller therefore keeps three parts separate: task-phase recognition, motion prediction and the distance limit.

The aim of this project is to make human–robot ceiling-panel installation practical by reducing unnecessary robot stops without reducing protection. This aim leads to one research question (RQ): *can a controller that knows the task phase keep people as safe as the fixed-zone control used in practice today, while keeping the task efficient?* Our main hypothesis (H1) is that the context-aware predictive controller is as safe as fixed-zone control while maintaining comparable task efficiency. Its safety rests on the same distance limit: the learned parts are not allowed to relax it.

The RQ cannot be answered by a single number, so it is broken into three supporting questions, each feeding one half of H1. (SQ1) Can the task phase be recognised accurately enough in real time to be useful? This is a precondition for any context-aware benefit. (SQ2) How do the controllers compare on the recorded safety and efficiency indicators, and how far can those recordings be trusted given the distance measurement and scripted movements? This tests H1 directly. (SQ3) How did participants experience working with the robot? This asks whether the system is acceptable to the people it is meant to help. Section VII-A returns to each question and states what the results answer.

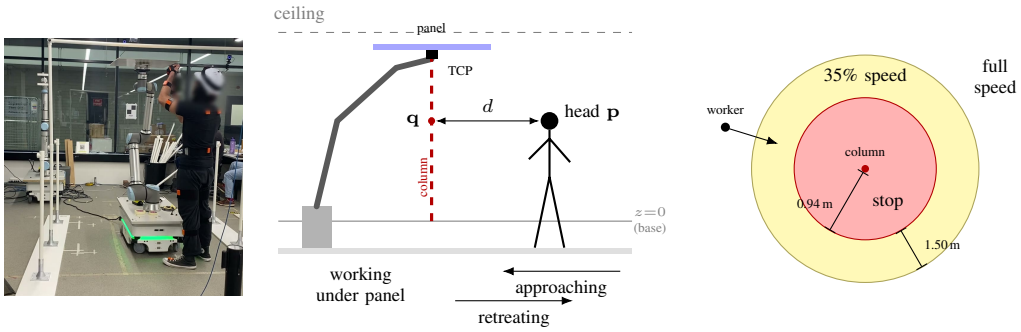
This work makes three contributions: (1) a physical ceiling-panel prototype in which task-phase recognition and intrusion prediction are kept separate, and both sit below a fixed speed limit; (2) a descriptive analysis of the distance inputs and robot motion recorded during the task; and (3) item-by-item questionnaire results, checked again using only complete response sets. The contribution is combining and testing these parts in a construction task, not a new SSM formula.

## II. Related Work

### A. Human–Robot Collaboration in Construction

Construction robotics is moving from fully automated machines toward systems that combine human skill with robot strength [1], [9]. Reviews of human–robot collaboration (HRC) in construction list safety, sensing, communication and task sharing as the main barriers to use on site [10], [11].

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(a) (b) (c)

Fig. 1. The application scenario. (a) Ceiling-panel installation in our lab: the robot holds the panel at ceiling height while the worker fixes it from below (frame from session video; face blurred). (b) Side view (schematic): the worker approaches, works under the panel and retreats; the robot is represented by a vertical column under the tool centre point (TCP), and $d$ is measured from the head $\mathbf{p}$ to the nearest column point $\mathbf{q}$ (later system version). (c) Plan view, radii to scale: the fixed controller's zones around the column and the speed allowed in each.

Panel installation is a useful test case because the person and the robot work around the same object, and stopping too often directly slows the work.

### B. Speed and Separation Monitoring

SSM needs more than a fixed safety radius around the robot. The required distance depends on how far the person and the robot move while the system senses, decides and brakes. It also includes margins for body parts that are not tracked and for sensor error [3], [4]. Byner et al. [5] set the robot's speed limit from the separation distance and direction of motion, and measured cycle performance in a physical machine-tending task. Karagiannis et al. [6] switch between safety-zone layouts depending on where the robot is, using an overhead safety camera, a safety-rated programmable logic controller (PLC) and the robot controller in a cell modelled on the car industry. Adaptive SSM and its effect on productivity have therefore already been studied, and they define the reactive baseline that task context must improve on.

Construction studies narrow the gap further. Xiao Lin et al. [7] combine worker, robot and material shapes in a digital twin that recalculates the protective distance as the scene changes, and test it in static and moving conflict scenarios. Their treatment of carried material matters for panel installation: once the robot carries a panel, a shape drawn around the robot alone no longer covers the space it occupies.

### C. Activity Recognition and Collaboration

A hidden Markov model (HMM) tracks a state that changes over time but cannot be observed directly. It uses a transition matrix, which gives how likely the state is to change, and an emission model, which gives the measurements each state tends to produce [12]. HMMs have been used to recognise activities from wearable sensors and to model human–robot tasks [13], [14].

In these studies the recognised states describe activities; whether hazards belong among them is discussed in Section III-D.

The closest construction study, by Xiao Lin et al. [8], combines a model of the work steps, robot behaviour learned from demonstration and local safety control in a virtual-reality wall-panel task with simulated sensors. Our study adds a physical ceiling-panel prototype with real robot data and participant questionnaires.

How smoothly people and robots work together is usually measured with timing, such as human idle time, robot idle time, time spent working in parallel and delay between actions [15]. Workload and trust questionnaires help explain results, but they cannot prove physical safety or replace measured robot response [16], [17]. This paper therefore links task recognition, measured control inputs and participant experience in one physical panel task. Table I compares these approaches with our prototype.

TABLE I
RELATED METHODS AND HOW THEY RELATE TO THIS STUDY.

| Study | Method and test setting | What it means for this study |
|---|---|---|
| Byner et al. [5] | Speed limit set continuously from distance and direction of motion; tested in a physical machine-tending cycle. | Reactive speed adjustment is the baseline; phase recognition must add value on top of it. |
| Karagiannis et al. [6] | Safety zones switch with robot position; overhead safety camera and safety PLC in a car-industry cell. | A software speed limit is not the same as a system built from safety-rated parts. |
| X. Lin et al. [7] | Protective distance updated from worker, robot and material shapes; construction conflict scenarios. | Panel and body shapes need checking beyond whether a point-based distance is calculated consistently. |
| X. Lin et al. [8] | Learned work-step model and local safety control; VR wall-panel task with simulated sensors. | Task-aware safety control already exists; our focus is real robot data and participant experience. |
| This study | Real-time HMM task phase and motion prediction below a fixed speed limit; physical ceiling-panel prototype. | Robot data and questionnaires from a construction-style task with a person in the loop. |

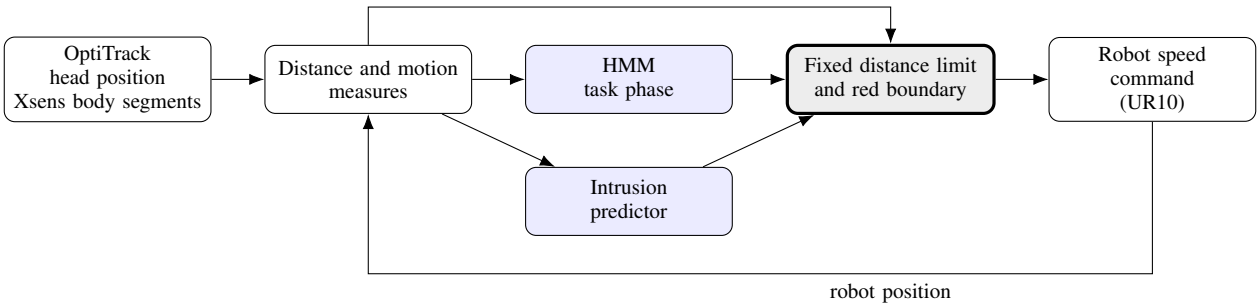



Fig. 2. How the prototype works. Motion capture and the robot's position give the distance and motion measures. The learned parts (blue) can only slow the robot further; they cannot raise its speed above the fixed distance limit (grey). Data collection for the experiment is shown separately in Fig. 4.

## III. System Design

### A. Overview

The controller (Fig. 2) has two inputs: from motion capture, the position of the worker's head relative to the robot and the head's velocity in the room frame; and from the robot controller, the position of the robot's tool. Its output is a single number between 0 and 1: the fraction of full speed the robot may use. This number is sent to the robot's speed override, and a value of zero requests a pause.

### B. Equipment and Sensing

The lab prototype (Fig. 3) uses a Universal Robots UR10 arm with a vacuum gripper that holds a lightweight stand-in for a ceiling panel. The arm lifts the panel to ceiling height, the participant walks in to work on it, and the arm then lowers it. Two separate systems track the participant.

*Xsens MVN* is a motion-capture suit. Seventeen small motion sensors, each with an accelerometer, gyroscope and magnetometer, are strapped to the body. Software combines their signals with a body model to give the position and orientation of 23 body segments, such as the pelvis, head, arms and legs, at up to 240 Hz [18], [19]. Because position is calculated by adding up small movements over time, it slowly drifts, and its starting point is set by a calibration pose rather than by the room. Here, Xsens provides the body-segment positions used for the body-distance input in the later version of the system.

*OptiTrack* is a camera-based motion-capture system. Infrared cameras around the cell (on the overhead frame in Fig. 3a) detect small reflective markers, and the Motive software locates each marker to about a millimetre in a fixed room frame [20]. A group of markers on the participant's cap is tracked as one object. OptiTrack therefore gives the head's position relative to the robot without drift, but says nothing about the rest of the body. Here, the OptiTrack head position is used to calculate the distance and closing speed that drive the phase model and the reactive limit.

The two systems' estimates are not merged; both feed the same processing step in the same format. The robot controller reports the tool centre point (TCP), which is the position of the gripper face, and the speed of each of the six joints. The control loop runs at about 60 Hz, and the actual time between samples is logged.

TABLE II
Symbols used in the controller equations.

| Symbol | Meaning |
|---|---|
| $d$ | Distance from tracked point to robot (m) |
| $v_{\mathrm{proj}}$, $a_{\mathrm{proj}}$ | Speed (m/s) and acceleration (m/s$^2$) toward the robot |
| $h_t$ | How directly the movement points at the robot |
| $z_t$ | Task phase at time $t$ (approaching, working, retreating) |
| $w_{ik}, \boldsymbol{\mu}_{ik}, \boldsymbol{\sigma}^2_{ik}$ | Weight, mean and variance of component $k$ in phase $i$ |
| $r_{\mathrm{red}}$ | Radius of the inner (red) boundary (m) |
| $\tau$ | Predicted time to reach the red boundary (s) |
| $S$ | Protective separation distance (m) |
| $T$ | Assumed time to react and stop (s) |
| $C$ | Margin for body parts not tracked (m) |
| $S_a$ | Margin for sensor error (m) |
| $u_{\mathrm{env}}, u_{\mathrm{context}}, u_{\mathrm{cmd}}$ | Speed fractions (0–1) from the distance limit, from the learned parts, and sent to the robot |

### C. Distance Measures and System Versions

The robot is represented by a vertical column, a simple stand-in for its shape (Fig. 1b). The column is a vertical line segment, with no radius, through the horizontal position of a robot reference point; it runs from the robot-base plane ($z = 0$) up to the height of that point. All positions are in the robot-base frame. For the tracked head position $\mathbf{p}$, the nearest point $\mathbf{q}$ on the column lies on the column at the head's height, limited to the column's height range. The controller calculates

$$d = \|\mathbf{q} - \mathbf{p}\|_2, \qquad v_{\mathrm{proj}} = \mathbf{v}^{\mathsf{T}}\mathbf{u}, \tag{1}$$

where $d$ is the head-to-robot distance in metres, $\mathbf{v}$ is the head velocity in the room frame (the least-squares slope of the last five samples; the robot's own motion is not subtracted), $\mathbf{u}$ is the unit vector pointing from the head toward the robot, and $v_{\mathrm{proj}}$ is the closing speed in m/s. A positive $v_{\mathrm{proj}}$ means the person is moving toward the robot. The heading alignment $h_t$, used below, measures how directly the person's movement points at the robot; it is not the direction the body faces. Table II lists all symbols.

Data were recorded with two system versions that place the column differently. In the earlier version the reference point was fixed on the robot base's vertical axis, 2.2 m high, so the column did not follow the arm. In the later version it is the live TCP, read at every step and used only if less



(a) (b) (c)

Fig. 3. Lab equipment. (a) The cell: the UR10 arm holding the panel stand-in at ceiling height, OptiTrack cameras on the overhead frame, and the operator desk. (b) A participant wearing the Xsens suit and OptiTrack cap, seen from behind while walking toward the robot. (c) The working phase: the participant works under the raised panel. Faces are blurred.

than 0.25 s old, so the column moves with the gripper and panel. The later version also streamed data automatically rather than on operator confirmation, and for control it used the 23 Xsens body segments, placed at the OptiTrack cap and taking the smallest distance and largest closing speed. The phase model used head measures in both. The versions are reported separately (Section V-B), not as an experimental condition.

### *D. Task Phase and Intrusion Prediction*

The model recognises three task phases: approaching, working and retreating. These labels describe what the person is doing, whereas a hazard is about being exposed to harm, and a person can trip or move quickly toward the robot during any phase. Treating "hazard" as a fourth phase would therefore mix up activity and danger. The HMM estimates only the task phase, and fast movement toward the robot is detected separately by the intrusion predictor below, so the distance limit can be tested on its own whatever the HMM decides. At each time step it uses four measures, $\mathbf{x}_t = [d_t, v_{\mathrm{proj},t}, \|\mathbf{v}_t\|, h_t]^{\mathsf{T}}$: distance, closing speed, overall speed and heading alignment. For each phase, the typical values of these measures are described by a mix of three Gaussian (bell-shaped) distributions:

$$p(\mathbf{x}_t \mid z_t = i) = \sum_{k=1}^{3} w_{ik}\mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_{ik}, \mathrm{diag}(\boldsymbol{\sigma}_{ik}^2)), \qquad (2)$$

where $z_t$ is the phase at time $t$, $w_{ik}$ is the weight of component $k$ in phase $i$ (the three weights add to one), and $\boldsymbol{\mu}_{ik}$ and $\boldsymbol{\sigma}_{ik}^2$ are that component's mean and variance for each measure. In real time the model can use only present and past data (the causal filter). For offline checking, the Viterbi method can also use later data, so results for the two are reported separately. Because the labelled phases are long and the model expects phases to persist, it changes phase slowly and may react late to short movements.

The intrusion predictor works independently of the HMM. It solves

$$d - r_{\mathrm{red}} - v_{\mathrm{proj}}\tau - \tfrac{1}{2}a_{\mathrm{proj}}\tau^2 = 0 \qquad (3)$$

for $\tau$, the time the person would take to reach the inner (red) boundary of radius $r_{\mathrm{red}}$ from the current distance $d$ if they kept their current closing speed $v_{\mathrm{proj}}$ and acceleration $a_{\mathrm{proj}}$. An intrusion is flagged when a risk score that rises as $\tau$ falls below 0.5 s passes its threshold (equivalent to $\tau \lesssim 0.58$ s) while the closing speed is at least 0.6 m/s. $a_{\mathrm{proj}}$ is the slope of the last three closing speeds, a derivative of a noisy derivative that reached about $\pm 30$ m/s$^2$ unclamped during development, far beyond a walking person. It is therefore capped at $\pm 4$ m/s$^2$ (about $0.4\,g$), and the 0.6-m/s gate stops sway near the boundary counting as an approach. Both are engineering settings chosen in simulation to suppress noise, not measured or standard values. In the later version the body-segment control input sets $a_{\mathrm{proj}} = 0$, so the cap had no effect there. The predictor detects movement toward the boundary, not intention.

### *E. Controllers and Speed Command*

Three controllers were compared. The *fixed* controller uses two zones around the column (Fig. 1c): inside the red radius it requests a stop, inside the yellow radius it limits the robot to 35% of full speed, and outside both it allows full speed. Leaving a zone requires clearing its boundary by 0.05 m, so the command does not flicker at the edge. The *reactive* controller also takes closing speed into account, using a simplified separation limit. The *predictive* controller adds phase-based caution and the intrusion predictor on top of the reactive limit. Its protective separation distance is

$$S = \max(0, v_{\mathrm{proj}})T + C + S_a, \qquad (4)$$

where $S$ is the distance the controller tries to keep; $T$ is the assumed time for the system to react and the robot to stop, so $v_{\mathrm{proj}}T$ is how much closer the person gets in that time; $C$ is a margin for body parts that the tracked point does not cover; and $S_a$ is a margin for sensor error. This is the ISO/TS 15066 separation formula without the term for the robot's own movement [3]. We used $T = 0.4$ s, $C = 0.2$ m and $S_a = 0.1$ m. Beyond $S$, the allowed speed rises gradually over 0.3 m. The red radius, 0.94 m, is the same formula evaluated at an assumed approach speed of 1.6 m/s ($1.6T + C + S_a$), and the yellow radius is 1.6 times that (1.50 m). These values were chosen for the lab cell; they are not measured stopping times or sensor errors of an installed system.

The speed sent to the robot is

$$u_{\mathrm{cmd}} = \min(u_{\mathrm{env}}, u_{\mathrm{context}}), \qquad 0 \le u_{\mathrm{cmd}} \le 1, \qquad (5)$$

where $u_{\mathrm{env}}$ is the speed fraction allowed by the reactive limit, $u_{\mathrm{context}}$ is the fraction suggested by the phase model and

TABLE III
LEAVE-ONE-OPERATOR-OUT PHASE-RECOGNITION VALIDATION ON THE DEVELOPMENT SET. VITERBI DECODING USES THE FULL SEQUENCE; THE CAUSAL FILTER USES PRESENT AND PAST OBSERVATIONS.

| Measure | Value |
|---|---|
| Development participants | 3 |
| Accepted trials | 32 |
| Labelled frames | 90487 |
| Offline Viterbi accuracy | 0.821 |
| Offline balanced accuracy | 0.757 |
| Causal-filter accuracy | 0.794 |
| Causal-filter balanced accuracy | 0.725 |
| Approaching recall, offline / causal | 0.693 / 0.684 |
| Working recall, offline / causal | 0.923 / 0.905 |
| Retreating recall, offline / causal | 0.654 / 0.586 |
| Offline fold P03 accuracy | 0.812 |
| Offline fold P04 accuracy | 0.817 |
| Offline fold P05 accuracy | 0.832 |

intrusion predictor, and $u_{\mathrm{cmd}}$ is the fraction sent to the robot. Taking the smaller of the two means the learned parts can only lower the speed, never raise it. Inside the red radius the robot is always asked to stop. If the head position, body data or robot position are missing or too old (more than 150 ms for OptiTrack or 0.25 s for the robot position), every controller requests a stop ($u_{\mathrm{cmd}} = 0$) before its own rules are applied. The command is sent through the robot's speed override and pause function. This is a research controller: its stop is not a safety-rated protective stop, and (5) limits the software request rather than certifying the installed system (Section VII-E).

The HMM's phase caution can only cap the speed at the reduced level, so it never stops the robot on its own. The intrusion predictor stops once the risk has exceeded its threshold for two consecutive samples and releases as soon as it falls back below the same threshold, with no hysteresis.

## IV. MODEL DEVELOPMENT AND VALIDATION

We trained the phase model before the participant study, using 32 labelled trials in which the three team members acted as operators (study IDs P03, eight trials; P04, 12; P05, 12). These contain 90,487 frames: 17,267 approaching, 53,609 working and 19,611 retreating. Labels were recorded live: the experimenter pressed a key at each real phase change to advance the dashboard's fixed task sequence, and each frame carries the current phase. *Approaching* runs from the start cue until the participant reaches the gripper with the panel or stands under the raised panel; *working* runs until the panel is gripped or, in the second cycle, a brief hazard movement is cued; *retreating* runs until the robot's lift or lowering is confirmed. Frames during robot motion and the hazard movement are excluded. All 32 trials follow the scripted order, but the labels were not reviewed afterwards or checked by a second labeller. We validated it by leaving one operator out at a time: the model was trained on two operators and tested on the third.

Two measures are reported. Accuracy is the share of frames whose predicted phase matches the label. Because most frames are "working", a model that favours that phase scores well on accuracy. Balanced accuracy averages the recall of the three phases, so each phase counts equally. Offline accuracy is 0.821 and real-time (causal) accuracy is 0.794 (Table III). Balanced accuracy is lower, especially for the causal filter (0.725), and recall is highest for the working phase. Because the controller runs online, the causal recalls are the relevant ones: 0.684 for approaching, 0.905 for working and 0.586 for retreating, against 0.693, 0.923 and 0.654 offline. Retreating loses most, as the filter is slow to leave the working phase. The small number of operators, the correlation between frames in the same trial, and the reuse of the same folds during development limit how far these values generalise.

*What this means for SQ1:* the phase can be read from body motion in real time, most reliably while the worker is under the panel, which is the phase in which unnecessary stops are most disruptive. The precondition for context-aware control is met; the weaker transition phases are where future data should go.

## V. EXPERIMENTAL METHOD

### A. Task and Experimental Design

The protocol has three blocks, A, B and C. The fixed, reactive and predictive controllers were assigned to the blocks in one of six rotated orders. Each block contains one clean trial, one distractor trial and one rapid-movement trial (Fig. 4). Each trial follows the panel sequence: walk in and load the panel, step back before the lift, work under the raised panel, and step back before it is lowered and unloaded. The trial types differ only during the lift, while the participant waits at the start marker. In a *clean* trial there is no cue. In a *distractor* trial, on the operator's "now", the participant makes one brisk sideways movement that does not close on the robot and returns. In a *rapid-movement* trial they make one brief move toward the robot along a marked direction and return at once (not a fall, trip or lunge). Both movements were demonstrated and practised first. A cue window marks the instruction, not the start of the movement.

### B. Participants and Analysed Records

The study ran under Monash University ethics approval (project 51642). The team recorded 12 people taking part, but because some sessions were restarted under a new P-number, that count cannot be rebuilt from the exported records alone. Study IDs therefore label recorded sessions, not confirmed individuals. In this paper a code such as P17 is a study ID, and T05 is a trial within it.

Because not every record could be linked to one person, the analysis is descriptive. The main data set is the 11 study IDs that have both intake and end forms and a known controller order. These IDs had 99 planned trial slots, for which 115 attempts were found: 108 completed and seven aborted. For each slot we used the latest completed trial whose files were readable and whose sample count matched its record. This gave 97 trials; two slots (P13 block A trial 1 and P22 block A trial 3) had no completed trial. Earlier repeats of a slot are kept in the audit record but not analysed. The 52 trials from P13–P22 were recorded with the earlier system version and the 45 from P31–P38 with the later one (Section III). The

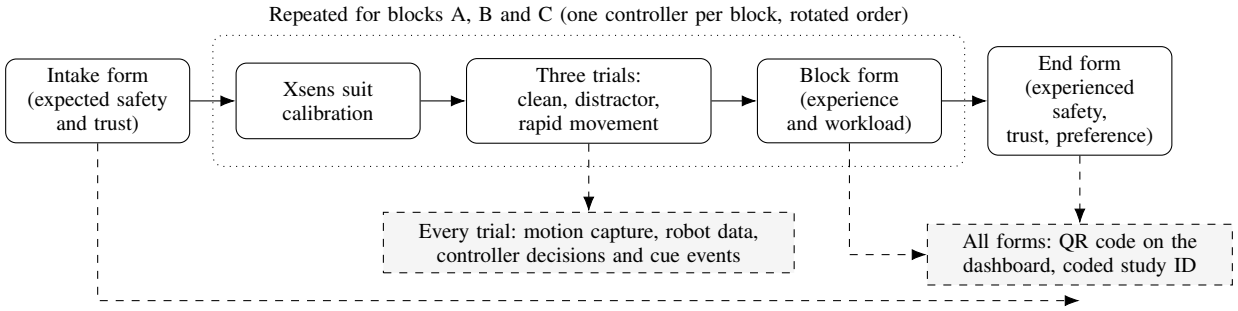



Fig. 4. One participant session. Solid arrows show the order of steps; dashed arrows show the data collected. The three controllers were assigned to blocks A, B and C in one of six rotated orders.

same 11 IDs gave 31 block questionnaires: nine IDs completed all three, and two are missing the last block. All 11 have matched intake and end forms. Robot data and questionnaires were analysed separately, and all values come directly from recorded samples and original form responses; no missing values were filled in.

### C. Robot Data Processing

The selection above gave 97 trials and 494,075 samples. File fingerprints, row counts and time totals were rechecked independently after the analysis. For time totals, each sample's value is held until the next sample, but only when the gap to the next sample is between 0 and 0.25 s. Longer gaps are reported separately and left out of both the total and the time base. A robot-data sample counts as fresh when the packet is available, all six joint speeds are valid and the data are less than 0.25 s old. The robot counts as moving when its fastest joint exceeds 0.005 rad/s; counts at 0.001 and 0.01 rad/s are kept as a check on this threshold.

To check the head distance, we recalculated it from the same sample's head position and the robot's live tool position. This checks internal consistency, not physical calibration. For each cue window we report the smallest distance and the largest closing speed, which need not occur at the same moment. Stop time includes intentional holds in the task, so it is not a count of false stops.

The recorded stop counts were checked for a known system issue: the speed command sometimes switched rapidly between stop and run. To separate this switching from real stops, every stop request (a change of the command from above zero to zero) was classed by duration, as *brief* (under 0.5 s) or *sustained* (0.5 s or longer), and by the controller rule logged with it: sensing-dropout hold, intrusion predictor, red boundary or envelope minimum. The same rules were applied to all three controllers, and the raw counts reproduce the earlier totals for every trial. As a check, stops were also joined when separated by less than 0.25, 0.5 or 1.0 s and counted only if they lasted at least 0.5 or 1.0 s. No video was used.

### D. Questionnaire Inclusion and Scoring

Pilot, link-test and qualification submissions were excluded. This left 18 intake forms, 31 block forms and 11 end forms, with 11, 11 and 9 block forms for A, B and C and no duplicates.

The block questionnaire uses five-point ratings. Three items run between two opposite words: anxious to relaxed, agitated to calm, and quiescent to surprised. Four items ask how much the participant agrees that they were confident the robot would stop, had to watch it constantly, felt it stopped more than needed, and found its speed changes made sense. Three items rate mental demand, physical demand and frustration. The workload items are adapted from NASA-TLX [16] and the confidence and monitoring items from Jian et al. [17]. Neither set is the full validated questionnaire, so each item is analysed on its own and no combined score is formed.

Block results are given as response counts and median [first, third quartile]. As a check, we repeat the summary using only the nine IDs with all three block forms, and count how many ratings went up, stayed the same or went down from A to C. These changes describe the order of the blocks; a higher rating is not always better. Results are grouped by block label, not by controller, because the assignment records alone do not prove which controller each questionnaire refers to. The intake and end forms are matched by study ID and ask about expected and then experienced safety and trust, from 1 (not at all) to 5 (completely). Because neither the number of independent people nor full controller exposure can be confirmed, we report no significance tests, confidence intervals or effect sizes, and we do not assume missing data are random.

## VI. Results

The results are presented in the order of the supporting questions: phase recognition (SQ1, Section IV), the recorded robot behaviour and how far it can be trusted (SQ2, below), and participants' experience (SQ3). Each subsection closes with what it means for H1.

### A. Robot Data

Each of the 11 study codes has eight or nine selected trials. Table IV summarises them by controller; these totals describe the records and are not estimates of a controller effect. The 97 trials contain 494,075 samples and 186.17 minutes of capture. 398 gaps are longer than 250 ms, totalling 281.65 s (2.52%); 10,888.35 s remain for time totals. 3244 samples are flagged as stale tracking, and 395 triggered the tracking-loss stop request.

TABLE IV
RECORDED BEHAVIOUR BY CONTROLLER (97 TRIALS). BRIEF STOP REQUESTS LAST LESS THAN 0.5 S.

| | Fixed | Reactive | Predictive |
|---|---|---|---|
| Trials | 32 | 32 | 33 |
| Clean / distractor / rapid | 10/11/11 | 10/11/11 | 11/11/11 |
| Median trial duration (s) | 112.5 | 113.6 | 109.0 |
| Median closest distance (m) | 0.08 | 0.06 | 0.08 |
| Time inside red boundary (%) | 28.5 | 31.3 | 32.0 |
| Stop requested inside red (%) | 100 | 100 | 100 |
| Median stop time (%) | 29.7 | 33.6 | 36.7 |
| All stop requests | 659 | 888 | 1417 |
| Brief: sensing dropout hold | 450 | 414 | 326 |
| Brief: intrusion predictor | 0 | 0 | 805 |
| Brief: other rules | 43 | 293 | 69 |
| Sustained: red boundary | 164 | 156 | 101 |
| Sustained: intrusion predictor | 0 | 0 | 104 |
| Sustained: other rules | 2 | 25 | 12 |
| Sustained stops per minute | 2.7 | 3.1 | 3.6 |

Totals are descriptive, not controller-effect estimates: order, participant identity and system version are not jointly controlled. Stop time includes intentional task holds. P13 A1 and P22 A3 have no completed capture.

Controller decisions exist in 494,023 samples, and every one is recorded as sent to the robot. This confirms that commands were recorded, not how the robot physically stopped.

The trials are close to the planned 11 per trial type per controller; the two missing slots leave 10 clean fixed and 10 clean reactive trials. Median stop time is 29.7%, 33.6% and 36.7% for fixed, reactive and predictive records, and the raw numbers of stop requests are 659, 888 and 1417. Because controller order, system version and person identity are not jointly controlled, these differences are descriptive. Stop time also includes intentional task holds.

The safety indicators were similar across controllers (Table IV). Whenever the measured distance was inside the 0.94-m red boundary, all three controllers requested a stop 100% of the time. The median closest distance per trial was 0.08, 0.06 and 0.08 m, and people spent 28.5%, 31.3% and 32.0% of observed time inside the red boundary, mostly during the loading, working and unloading approaches the task requires. Median trial duration was 112.5, 113.6 and 109.0 s, so predictive trials were not slower overall, although the robot spent more of that time stopped.

Most stop requests were brief: 493 of 659 for the fixed controller, 707 of 888 for the reactive controller and 1200 of 1417 for the predictive controller lasted less than 0.5 s (Table IV). The sensing-dropout hold shared by all three controllers produced 450, 414 and 326 of them, and the intrusion predictor, used only by the predictive controller, produced 805 brief and 104 sustained requests. Sustained stops were closer: 166, 181 and 217 (2.7, 3.1 and 3.6 per minute), mostly red-boundary stops during the approaches the task requires; joining stops under 0.5 s apart and keeping those of at least 1 s gave the same order (140, 153 and 187).

In the 33 later trials, the log also records what all three controllers would have requested at each sample from the same inputs. On these identical inputs the predictive controller started a stop 1.89 times as often as the reactive controller (median per trial; interquartile range 1.61–2.43), so the switching comes from the controller itself rather than from participants or order (these open-loop values do not show how the person or robot would then have moved). The switching also grew over the study: predictive trials logged a median of about 49 Hz on 9–10 September and 38 Hz on 16–17 September, while brief intrusion-predictor stops rose from about 6 to 20 per minute (rank correlation $-0.72$, $n = 33$). Date, data rate and system version changed together, so their effects cannot be separated. Fig. 6 shows the switching in one trial.

Across the scheduled cue windows, fresh robot data show joint motion for 1712.80 s of observed time, confirming that the robot moved during the labelled event windows. Of 94,031 fresh samples in these windows, 79,356, 77,862 and 77,492 show motion at 0.001, 0.005 and 0.01 rad/s, so the result does not depend on the chosen threshold.

*What this means for H1:* on the recorded indicators, the safety half of H1 holds, since no controller let a person inside the red boundary without a stop request and close approaches were alike. The efficiency half holds only in part: the work took no longer, but the predictive robot paused more, and the same-input comparison shows the pauses come from the controller's own design rather than from who was in the task.

### B. Distance Inputs and Cued Movements

In the early records, the median difference between the recorded head distance and the same-sample recalculation (95th percentile per trial) is 0.481 m (Fig. 5a). In the later records it is 0.000 m at the reported precision. The early difference arises because the recorded distance was measured to the column on the robot-base axis, whereas the recalculation uses the column under the live TCP, so it grows as the arm carries the panel away from the base axis. The early distance inputs therefore describe the controller's view, not the true head-to-panel clearance. The zero difference shows that the later logging is arithmetically consistent, not that it is physically calibrated or safety-rated.

The wide spread and overlap of rapid and distractor cue windows (Fig. 5b) show that the scripted cue name does not decide the movement that was actually measured. These values are controller inputs, not checked physical clearances, so the measured distance and closing speed, not the cue name, must guide interpretation. Figure 6 shows a typical later record. Because the cue marks the instruction rather than the start of the movement, the delay between cue and robot response is not reported as a protective response time.

*What this means for H1:* these checks decide how firmly the safety half of H1 can be stated. In the later system version the recorded distances are internally consistent, so the controller comparisons rest on sound inputs; in the earlier version they describe what the controller saw rather than true clearance. The safety result is therefore a comparison of controller decisions, not yet a measured physical safety margin.

### C. Reported Experience

Block responses lean toward relaxed and calm, confidence that the robot would stop, and low demand and frustration (Table V). Median confidence that the robot would stop is

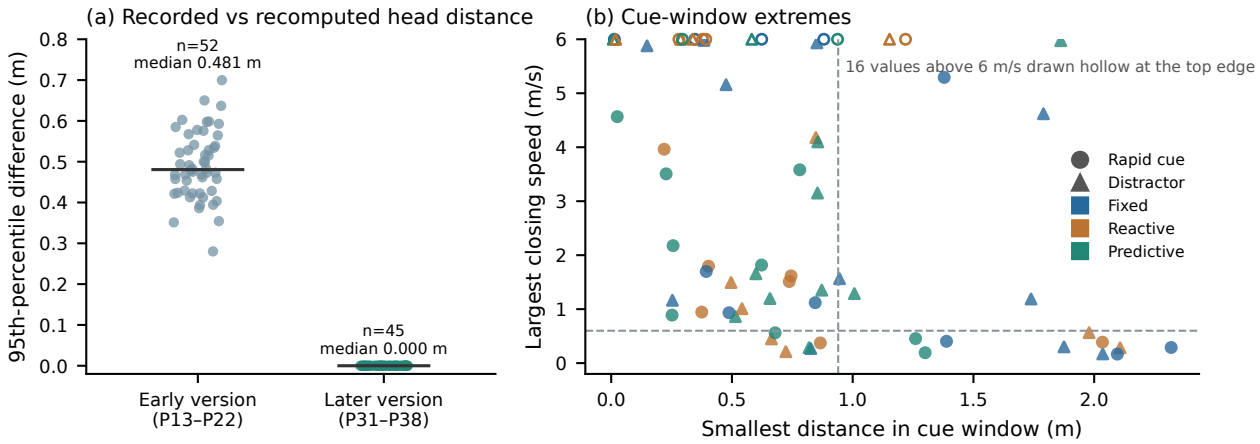



Fig. 5. Measurement checks, one mark per trial. (a) 95th-percentile difference between the recorded head distance and a recalculation from the same sample, by system version. (b) For each distractor or rapid cue window, the smallest distance (horizontal) against the largest closing speed (vertical). Dashed lines show the 0.94-m inner boundary and the 0.6-m/s closing threshold. Values above 6 m/s are physically implausible input spikes and are drawn hollow at the top edge. The two extremes need not occur at the same moment.

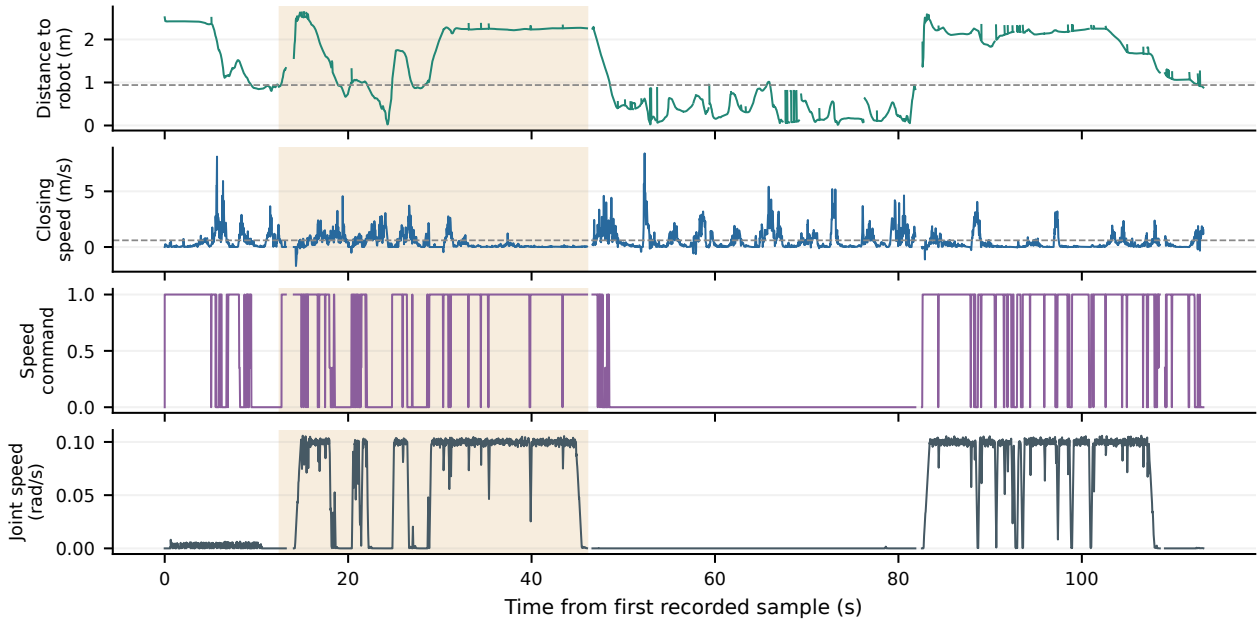



Fig. 6. A typical later trial (P31 T10, predictive controller): distance to the robot, closing speed, speed command and fastest joint speed. Shading marks the cue window; dashed lines are the configured thresholds; gaps over 250 ms break the lines. Note the rapid switching of the speed command.

five in every block, while median agreement that it stopped more than needed is three in A and two in B and C. This is a perception, not an objective count of false stops, and responses are clustered within study IDs.

Using only the nine complete response sets keeps 29 of the 30 block medians unchanged; the exception is constant monitoring in B, which changes from two to one. The paired A–C counts show variation that the medians hide (Table VI). Five relaxed ratings increase and four stay the same. Unnecessary-stopping ratings increase twice, decrease twice and stay the same five times despite the lower median in C, and surprise decreases for three codes, increases for two and is unchanged for four, so lower medians do not mean that everyone improved.

Across the 11 matched records, the mean safety rating was 3.82/5 before and 4.45/5 after the session: eight ratings increased, one was unchanged and two decreased. Mean trust was 4.00 before and 4.18 after: five increased, three were unchanged and three decreased (Table VII). The intake form asks about expected and the end form about experienced safety and trust, and individual linkage is incomplete, which limits how these changes can be read.

In the end forms, block C was chosen as safest seven times, compared with two each for A and B. For a full work shift, six chose C, three B and two A. Choices for least safe were spread across all blocks (A three, B four, C four). Because the link between block label and controller is not fully established, C is not read as the predictive controller. Four participants noticed changes between blocks and seven did not; their comments refer to stopping or slowing sensitivity and lift speed. The

TABLE V
BLOCK QUESTIONNAIRE ITEMS: MEDIAN [FIRST, THIRD QUARTILE]. A, B AND C ARE REPORTED BLOCK LABELS, NOT POOLED CONTROLLER CONDITIONS. ALL RATINGS USE 1–5.

| Item | A ($n = 11$) | B ($n = 11$) | C ($n = 9$) |
|---|---|---|---|
| Anxious to relaxed | 4 [3.5, 4.5] | 5 [4.0, 5.0] | 5 [4.0, 5.0] |
| Agitated to calm | 4 [4.0, 5.0] | 5 [4.0, 5.0] | 5 [4.0, 5.0] |
| Quiescent to surprised | 3 [1.0, 3.0] | 3 [1.0, 4.0] | 1 [1.0, 3.0] |
| Confident robot would stop | 5 [4.0, 5.0] | 5 [4.0, 5.0] | 5 [4.0, 5.0] |
| Had to watch constantly | 2 [1.0, 3.0] | 2 [1.0, 3.0] | 1 [1.0, 3.0] |
| Stopped more than needed | 3 [2.0, 4.0] | 2 [1.0, 3.5] | 2 [2.0, 3.0] |
| Speed changes made sense | 5 [3.5, 5.0] | 5 [4.0, 5.0] | 4 [4.0, 5.0] |
| Mental demand | 1 [1.0, 3.0] | 1 [1.0, 2.0] | 1 [1.0, 2.0] |
| Physical demand | 1 [1.0, 3.0] | 1 [1.0, 2.5] | 1 [1.0, 3.0] |
| Frustration | 1 [1.0, 3.0] | 1 [1.0, 2.0] | 1 [1.0, 1.0] |

Higher means the right-hand semantic anchor for the first three items, stronger agreement for the next four, and greater demand/frustration for the final three. Responses are clustered within 11 study IDs; individual-level linkage is incomplete. No trust, safety or workload composite is used.

TABLE VI
COMPLETE RESPONSE SETS: BLOCK MEDIANS AND WITHIN-CODE CHANGES ($n = 9$ CODES).

| Item | A/B/C | A to C: up/same/down |
|---|---|---|
| Anxious to relaxed | 4/5/5 | 5/4/0 |
| Agitated to calm | 4/5/5 | 3/6/0 |
| Quiescent to surprised | 3/3/1 | 2/4/3 |
| Confident robot would stop | 5/5/5 | 3/4/2 |
| Had to watch constantly | 2/1/1 | 1/5/3 |
| Stopped more than needed | 3/2/2 | 2/5/2 |
| Speed changes made sense | 5/5/4 | 1/5/3 |
| Mental demand | 1/1/1 | 2/6/1 |
| Physical demand | 1/1/1 | 1/6/2 |
| Frustration | 1/1/1 | 1/5/3 |

Counts refer to the raw rating direction, not uniformly to improvement. A–C changes describe block sequence, not a controller effect. The same nine codes supply every row.

TABLE VII
PAIRED GLOBAL RATINGS FOR 11 CODES WITH INTAKE AND END RESPONSES. FIVE-POINT ITEMS; DESCRIPTIVE CODE-LEVEL CHANGES ONLY.

| Item | Before | After | $\Delta$ | Up/same/down |
|---|---|---|---|---|
| Felt safe | 3.82 | 4.45 | 0.64 | 8/1/2 |
| Trust in distance | 4.00 | 4.18 | 0.18 | 5/3/3 |

Before/after wording changes from expected to experienced safety/trust. Means summarize coded responses; neither causation nor an interval-scale measurement model is assumed.

median rating for how much the suit restricted movement was two (interquartile range 1.5–3), and no end form reported risk or discomfort.

*What this means for SQ3:* people were comfortable working beside the robot under all three controllers and came away trusting it more than they expected, so the extra caution of the predictive controller did not make the experience worse. The questionnaires cannot, however, single out one controller as better.

## VII. DISCUSSION

### A. Answering the Research Question

Table VIII links each finding back to the question it answers and to the project aim.

Taken together, the findings answer the research question in two halves. A controller that knows the task phase can be built so that it is never less cautious than fixed-zone control, and in this study it was not less safe on any recorded indicator. It did not yet keep the task more efficient: it paused more, for a reason that lies in one component rather than in the idea of task-aware control.

*Significance for industry.* For construction firms and robot integrators, the main lesson is architectural. Task recognition and motion prediction can sit on top of an unchanged, standard distance limit, so a safety assessor still only has to validate that limit, while the learned parts add caution and context. This lowers the barrier to trying smarter collaboration on site. The study also warns that adding prediction is not automatically more productive: an over-sensitive predictor turns into frequent pauses, which on a real site would cost time and frustrate workers. Before such systems are deployed, firms will need measured end-to-end stopping response, geometry that covers the panel and tool, and productivity measured over full shifts, which are exactly the measurements this study identifies.

### B. Context-Aware Control in the Panel Task

Keeping the task phase and motion prediction below a fixed software limit separates two questions: whether a movement belongs to a normal task phase, and whether it is approaching the boundary. A wrong HMM phase cannot raise the speed above the distance limit, although geometry and timing errors still affect that limit.

Phase recognition is strongest in the long working phase, and because real-time balanced accuracy is lower than overall accuracy, phase accuracy alone does not show practical benefit. The predictive controller's higher raw stop count is mainly a switching effect: 805 of its 1417 stop requests were brief requests from the intrusion predictor, and the HMM caused none. The predictor stops after two samples above its threshold and releases as soon as the risk drops, so a noisy closing speed near the threshold flips the command between stop and run; the lower, less even data rate on later days is consistent with the rise in switching. A release delay or hysteresis should remove most of this without touching the distance limit. A minimum stop time is not a free fix: replaying the predictive commands

TABLE VIII
FROM RESULTS TO RESEARCH QUESTION AND AIM

| Question | What we found | Answer | Meaning for the aim |
|---|---|---|---|
| SQ1 Phase recognition | The phase is recognised in real time on unseen operators, best while working under the panel. | Yes: task context can be read reliably enough to drive a controller. | The building block for fewer unnecessary stops exists. |
| SQ2 Safety | Every controller requested a stop whenever a person was inside the inner boundary; close approaches and time near the robot were alike. | No sign that task awareness reduced safety (safety half of H1 supported). | Context can be added without weakening protection, because the learned parts can only slow the robot. |
| SQ2 Efficiency | Tasks took as long, but the predictive robot paused more, mainly from brief on–off switching of its approach predictor. | Comparable duration, more interruption (efficiency half of H1 only partly supported). | The aim of fewer stops is not yet met; the cause is identified and fixable. |
| SQ3 Experience | Participants trusted the robot to stop, reported low workload and felt safer afterwards. | The system is acceptable to users; no controller is singled out. | Workers are likely to accept a cautious, task-aware robot. |
| RQ / H1 | | Partly supported: as safe on the measured indicators, not yet more efficient. | Feasible and safe in design; efficiency gain still to be shown. |

with a 1-s minimum cut stop requests from 1417 to 420 but raised stopped time from 1351 to 2064 s.

Equation (5) also limits the expected benefit: on the same inputs the predictive controller can never request a higher speed than the reactive one, so any benefit must come from the timing of intervention or from how the person and robot then coordinate, which neither phase accuracy nor replay on recorded movement can show.

Against H1, the results split into two parts. On safety, we found no sign that the predictive controller was less safe than fixed-zone control: a stop was requested every time the measured distance was inside the red boundary, closest approaches and time inside it were similar, the learned parts can only lower the speed, and confidence that the robot would stop was high in every block. On efficiency, trials took about the same time (median 109.0 against 112.5 s), but the robot was stopped for more of each predictive trial (36.7% against 29.7%), with more sustained stops (3.6 against 2.7 per minute). H1 is therefore partly supported, and because participant, order and system version were not controlled, it remains inconclusive rather than confirmed or rejected. Fixing the predictor's switching is the clearest route to closing the efficiency gap.

### C. Measured Behaviour and Reported Experience

The difference between the two system versions strongly affects how the distance input can be read, and the gap between scripted and measured movements shows that event labels cannot replace measured exposure. Both support validating the whole chain from sensing to stopping [4].

Participants reported confidence that the robot would stop, low demand and little frustration, and felt safer after the session than they expected to. Novelty, supervision, repeated exposure and expectations could all contribute, so this cannot be credited to the predictive controller. Ceiling and floor effects leave little room to tell controllers apart, the complete-response check does not remove bias from missing data, and the preference for block C does not rank controllers because block labels rotate. Agreement that the robot stopped more than needed reflects perceived interruption, not an objective false-stop rate.

### D. Safety and Efficiency

Research on adaptive SSM motivates measuring robot availability and cycle performance [5], [6]. This study builds what is needed to measure them but does not show an advantage. Stop time includes intentional holds and capture time includes operator actions, so they do not establish false stops, task efficiency or fluency [15]. Formally showing equal safety would need a justified equivalence margin, validated protective geometry and measured end-to-end stopping response. All planned measures are reported: distance and stops per controller, trial duration in place of completion time, safety and trust as intake and end ratings, comfort through the relaxed and calm items, and predictability through the speed-change item.

### E. Limitations

The sample is small and was recruited for convenience. Because sessions were restarted, not every study ID can be linked to one person, so the 11 study codes cannot be treated as 11 independent people. Two trial slots lack a completed capture and two questionnaire sets lack block C, and missing data may relate to technical problems or task execution. The system version changed during collection, and the Xsens data rate fell on later days. Head points and body-segment positions do not cover the full body, tool and panel, and agreement with a calculation does not prove physical calibration. The controller is a research prototype: the separation limit is simplified, the command uses the ordinary robot speed override rather than a safety-rated stop, and the end-to-end response was not measured, so the results do not support a certified safety claim. Cued movements, operator pacing, visible robot motion, learning and fatigue may affect both behaviour and ratings. The phase model was built from three operators and may not generalise to construction workers.

### F. Future Work

Future tests should keep one system version, check its sensor error over the full moving volume, add hysteresis to the intrusion predictor, synchronise body movement, controller output and robot braking to measure response times, and use a study ID that follows each person across restarted sessions.

## VIII. Conclusion

This project aimed to make human–robot ceiling-panel installation more practical by cutting unnecessary robot stops without reducing protection, and asked whether a controller that knows the task phase can be as safe as fixed-zone control while keeping the task efficient. We built and tested a physical prototype in which phase recognition and fast-approach prediction sit below a fixed speed limit, so the learned parts can only make the robot more cautious.

The answer is a qualified yes on safety and not yet on efficiency. The task phase can be recognised in real time on people the model has never seen; every controller stopped the robot whenever someone came too close; and participants trusted the robot and felt safer after working with it. Tasks took no longer, but the predictive robot paused more, because its approach predictor switched on and off on noisy readings. H1 is therefore partly supported.

For industry, this shows that task awareness can be layered over an unchanged, standard safety limit without weakening it, which makes smarter collaboration easier to assess and adopt, but that its productivity benefit depends on careful tuning. The next steps follow directly: smooth the predictor with hysteresis, keep one validated distance measurement, measure the robot's real stopping response, and follow each participant across sessions, so that the efficiency half of the question can be answered conclusively.

## Data Availability

Analysis scripts, derived tables and figure-generation code accompany the manuscript. Participant-level data require controlled access. All reported summaries are calculated from recorded observations without imputation or simulated values.

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## AI Assistance Declaration

AI tools assisted with data-processing and analysis code, figure and table generation, and drafting and editing of the manuscript text.

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